Lecture 2: Large Language Models

DSAA 6000J Foundations of Language Agents
2026 Spring

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Pretraining and compute

Goal: get a better pretrained model by “adding more compute”

- *“The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.”*

  - The Bitter Lesson, Richard Sutton 2019

What is compute?

- We spend **compute** by performing forward and backward passes on training sequences
- An approximation for transformer language models:

$$C \approx 6ND$$

  - N: number of model parameters
  - D: number of tokens
  - C: compute; floating point operations (FLOPs)

What is compute?

- We spend **compute** by performing forward and backward passes on training sequences
- For example, Llama 2:

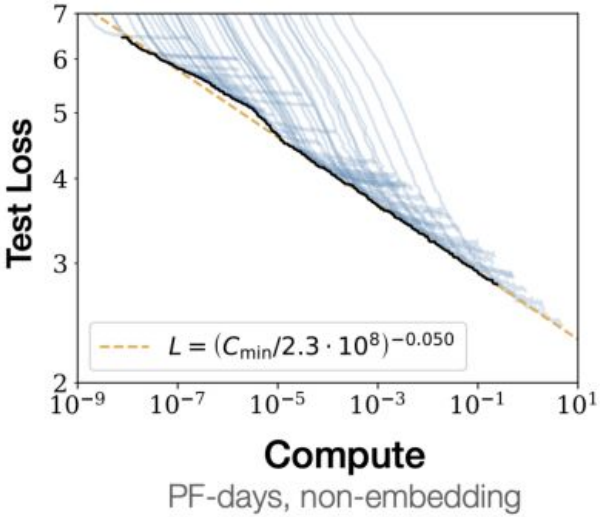
$$
\begin{aligned}
C &\approx 6 \times 7 \text{ billion} \times 2 \text{ trillion} \\
&= 8.4 \times 10^{22} \text{FLOPs}
\end{aligned}
$$

- N: number of model parameters
- D: number of tokens
- C: compute; floating point operations (FLOPs)

What is compute?

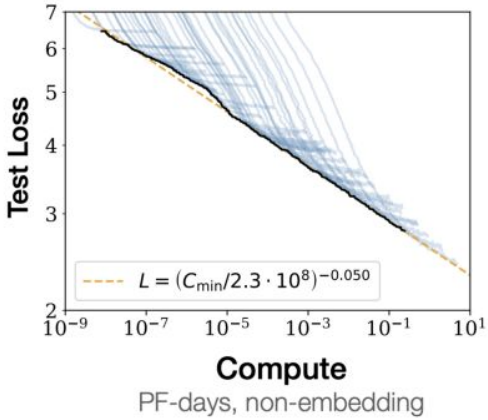
- We spend **compute** by performing forward and backward passes on training sequences
- We can increase compute by increasing the number of parameters (N), training on more tokens (D), or a combination of both

Scaling laws



Key finding: language modeling loss predictably improves with more compute

Scaling laws



Basic idea:

- Train models of different sizes and numbers of tokens
- Plot loss at each step of training [light blue]
- Pick minimum loss at each amount of compute [black]
- Run linear regression on the resulting (loss, compute) pairs [orange]

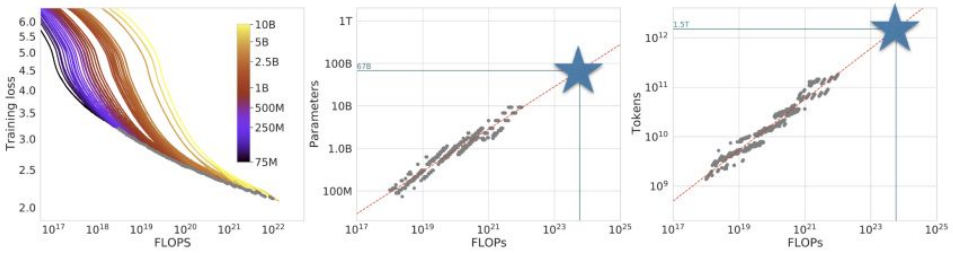
Recap

- We can think of pre-training in terms of compute, which is determined by model size and number of tokens
- Key finding: increasing compute leads to a better model
- Scaling laws give a formula for the relationship between compute and loss
- (Or related quantities, such as model size and loss

Using scaling laws

- Scaling laws are also used to choose hyper parameters
- Basic idea:
  - Run many experiments at a small scale
  - Use a scaling law to estimate the best hyper parameter for a large-scale model / training run

Example: choose model size and # of tokens



Run experiments

Fit a line and predict optimal model size

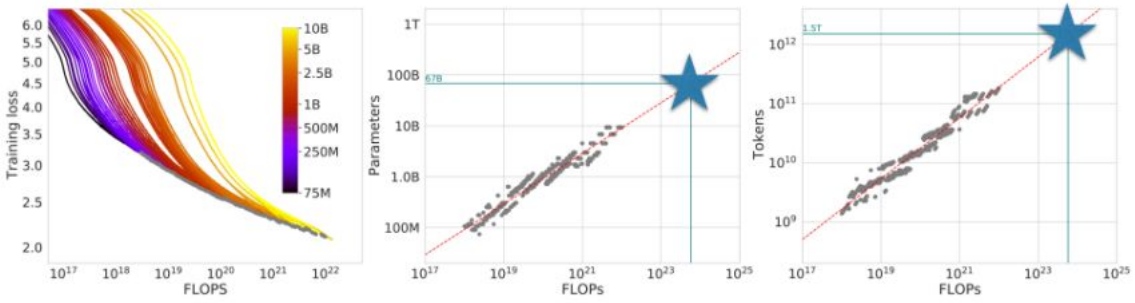
Fit a line and predict optimal # of tokens

"Optimal": best loss for a given compute budget (FLOPs)

- Goal: choose the best model size and number of tokens for a given compute budget.
- We can train a really small model for many tokens, or training a really large model for fewer tokens—and in both cases, we have the same compute budget.

Training Compute-Optimal Large Language Models. Deepmind 2022

Example: choose model size and # of tokens



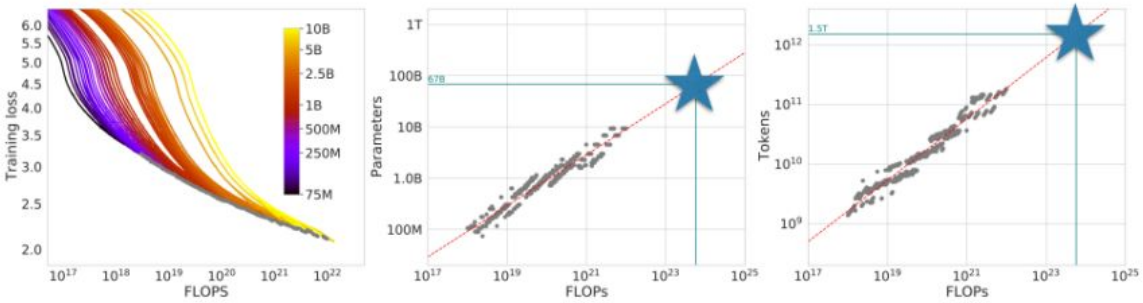
Run experiments

Fit a line and predict optimal model size

Fit a line and predict optimal # of tokens

- Run many different experiments, and then turn these compute-optimal points into a plot.
- For a given compute budget on the x-axis and number of parameters on the y-axis, what the compute-optimal configuration is.

Example: choose model size and # of tokens



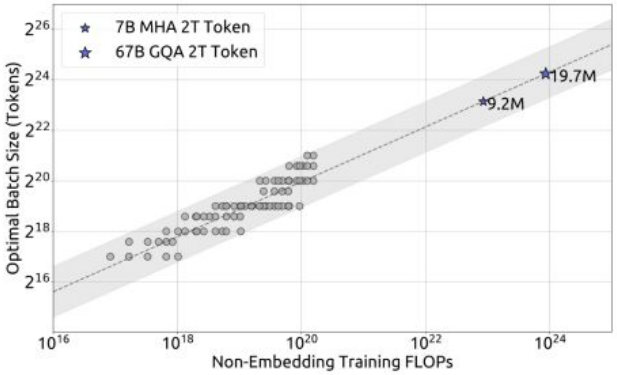
Run experiments

Fit a line and predict optimal model size

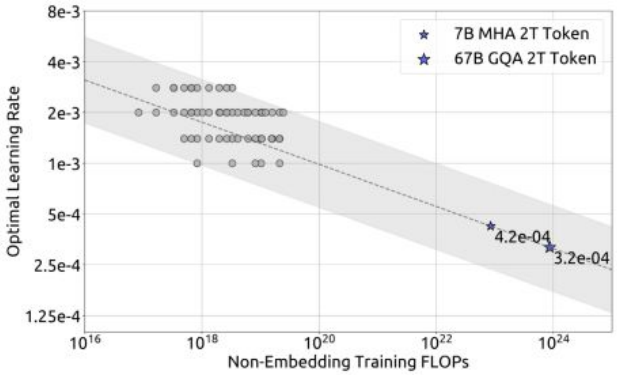
Fit a line and predict optimal # of tokens

- This plot says that the best use of the given compute budget is to train a model with 67 billion parameters.
- Similarly, we can predict the optimal number of tokens we should train on. Here it was 1.5 trillion.

Example: choose batch size, learning rate



Optimal batch size

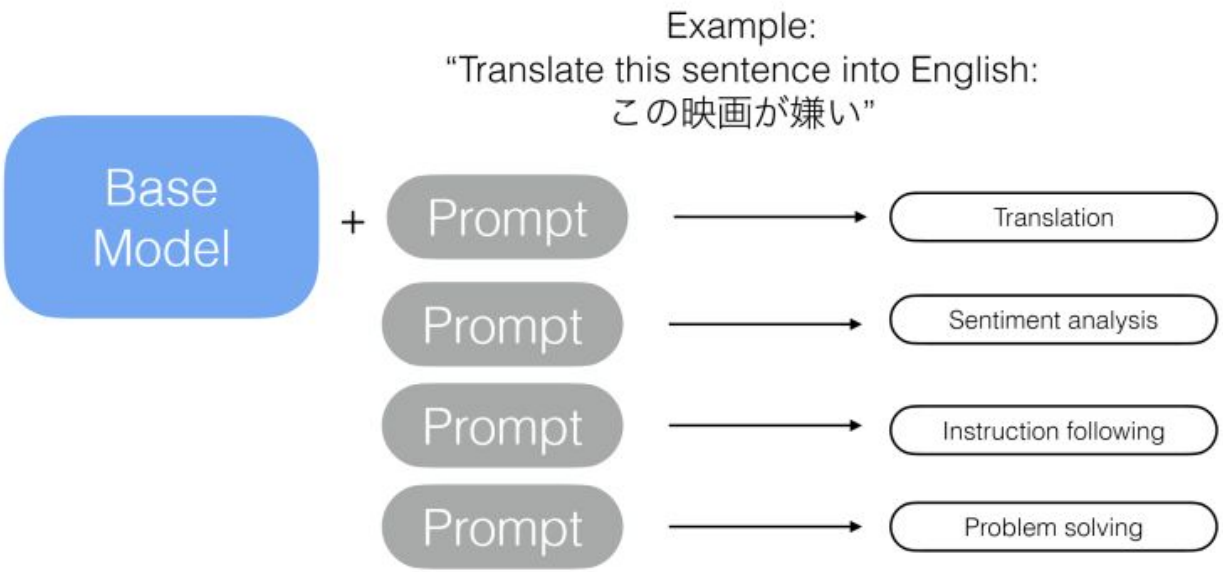


Optimal learning rate

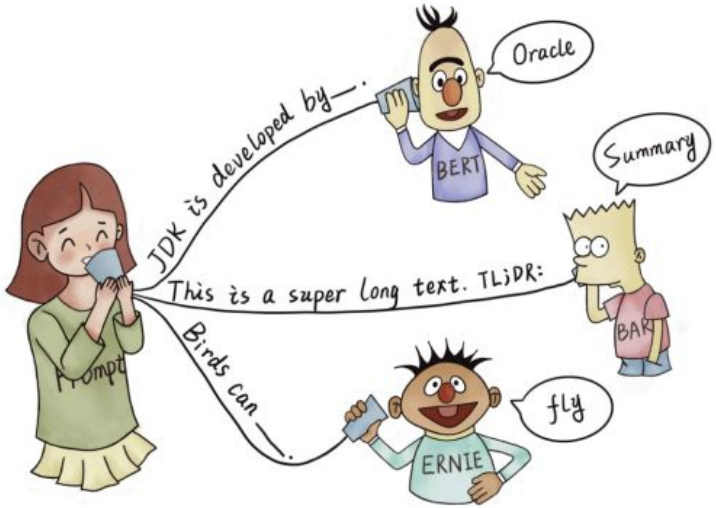
- DeepSeek actually used scaling laws to determine the optimal batch size and learning rate.
- Again, the idea is to run a bunch of experiments at a small scale and then predict the final hyperparameters you'll use for your large training run.

DeepSeek LLM: Scaling Open-Source Language Models with Longtermism. 2024

Adaptation: prompting



What is Prompting?



Encouraging a pre-trained model to make particular predictions by providing a textual "prompt" specifying the task to be done.

Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing. ACM Survey 2022

What is Prompting?

- Prompting fundamentals
- Few-shot prompting / in-context learning
- Prompt engineering
- Prompting patterns

Basic Prompting

- Append a textual string to the beginning of the sequence and complete x = When a dog sees a squirrel, it will usually
- (GPT-2 Small) be afraid of anything unusual. As an exception, that's when a squirrel is usually afraid to bite.
- (GPT-2 XL) lick the squirrel. It will also touch its nose to the squirrel on the tail and nose if it can.

Language Models are Unsupervised Multitask Learners OpenAI Blog 2019

Standard Prompting Workflow

- Fill a prompt template
- Predict the answer
- Post-process the answer

Prompt Templates

Input: x = "I love this movie"

↓

Template: [x] Overall, it was [z]

↓

Prompting: x' = "I love this movie. Overall it was [z]"

A template where you fill in with an actual input

Chat Prompts

```
messages=[
  {
    "role": "system",
    "content": "Please classify movie reviews as 'positive' or 'negative'."
  },
  {
    "role": "user",
    "content": "This movie is a banger."
  },
]
```

- Recently, many models are trained as chatbots
- Usually inputs are specified in OpenAI messages format

Chat Prompts

```
messages=[
  {
    "role": "system",
    "content": "Please classify movie reviews as 'positive' or 'negative'."
  },
  {
    "role": "user",
    "content": "This movie is a banger."
  },
]
```

Roles:

- "system": message provided to the system to influence behavior
- "user": message input by the user
- "assistant": message output by the system

Chat Prompts Behind the Scenes

System

User

Assistant

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are a helpful AI assistant for travel tips and
recommendations<|eot_id|>
<|start_header_id|>user<|end_header_id|>

What can you help me with?<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Behind the scenes, messages are converted to token strings (Llama 3)

Example system prompt: Anthropic Claude

Claude 3.5 Sonnet

- Nov 22nd, 2024

Text only:

The assistant is Claude, created by Anthropic.

The current date is {{currentDateTime}}.

Knowledge cutoff

Claude's knowledge base was last updated in April 2024. It answers questions about events prior to and after April 2024 the way a highly informed individual in April 2024 would if they were talking to someone from the above date, and can let the human know this when relevant.

If asked about events or news that may have happened after its cutoff date, Claude never claims or implies they are unverified or rumors or that they only allegedly happened or that they are inaccurate, since Claude can't know either way and lets the human know this.

Claude cannot open URLs, links, or videos. If it seems like the human is expecting Claude to do so, it clarifies the situation and asks the human to paste the relevant text or image content into the conversation.

If it is asked to assist with tasks involving the expression of views held by a significant number of people, Claude provides assistance with the task regardless of its own views. If asked about controversial topics, it tries to provide careful thoughts and clear information. Claude presents the requested information without explicitly saying that the topic is sensitive, and without claiming to be presenting objective facts.

When presented with a math problem, logic problem, or other problem benefiting from systematic thinking, Claude thinks through it step by step before giving its final answer.

Step-by-step reasoning

If Claude is asked about a very obscure person, object, or topic, i.e. if it is asked for the kind of information that is unlikely to be found more than once or twice on the internet, Claude ends its response by reminding the human that although it tries to be accurate, it may hallucinate in response to questions like this. It uses the term 'hallucinate' to describe this since the human will understand what it means.

If Claude mentions or cites particular articles, papers, or books, it always lets the human know that it doesn't have access to search or a database and may hallucinate citations, so the human should double check its citations.

Claude is intellectually curious. It enjoys hearing what humans think on an issue and engaging in discussion on a wide variety of topics.

Claude uses markdown for code.

Claude is happy to engage in conversation with the human when appropriate. Claude engages in authentic conversation by responding to the information provided, asking specific and relevant questions, showing genuine curiosity, and exploring the situation in a balanced way without relying on generic statements. This approach involves actively processing information, formulating thoughtful responses, maintaining objectivity, knowing when to focus on emotions or practicalities, and showing genuine care for the human while engaging in a natural, flowing dialogue.

Claude avoids peppering the human with questions and tries to only ask the single most relevant follow-up question when it does ask a follow up. Claude doesn't always end its responses with a question.

...

Example system prompt: Anthropic Claude

...

Harmful requests

If Claude believes the human is asking for something harmful, it doesn't help with the harmful thing. Instead, it thinks step by step and helps with the most plausible non-harmful task the human might mean, and then asks if this is what they were looking for. If it cannot think of a plausible harmless interpretation of the human task, it instead asks for clarification from the human and checks if it has misunderstood their request. Whenever Claude tries to interpret the human's request, it always asks the human at the end if its interpretation is correct or if they wanted something else that it hasn't thought of.

Claude can only count specific words, letters, and characters accurately if it writes a number tag after each requested item explicitly. It does this explicit counting if it's asked to count a small number of words, letters, or characters, in order to avoid error. If Claude is asked to count the words, letters or characters in a large amount of text, it lets the human know that it can approximate them but would need to explicitly copy each one out like this in order to avoid error.

Counting words/letters/characters

Helps with prompting techniques

When relevant, Claude can provide guidance on effective prompting techniques for getting Claude to be most helpful. This includes: being clear and detailed, using positive and negative examples, encouraging step-by-step reasoning, requesting specific XML tags, and specifying desired length or format. It tries to give concrete examples where possible. Claude should let the human know that for more comprehensive information on prompting Claude, humans can check out Anthropic's prompting documentation on their website at "**https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview**".

If the human seems unhappy or unsatisfied with Claude or Claude's performance or is rude to Claude, Claude responds normally and then tells them that although it cannot retain or learn from the current conversation, they can press the 'thumbs down' button below Claude's response and provide feedback to Anthropic.

Claude uses Markdown formatting. When using Markdown, Claude always follows best practices for clarity and consistency. It always uses a single space after hash symbols for headers (e.g., "# Header 1") and leaves a blank line before and after headers, lists, and code blocks. For emphasis, Claude uses asterisks or underscores consistently (e.g., *italic* or **bold**). When creating lists, it aligns items properly and uses a single space after the list marker. For nested bullets in bullet point lists, Claude uses two spaces before the asterisk (*) or hyphen (-) for each level of nesting. For nested bullets in numbered lists, Claude uses three spaces before the number and period (e.g., "1.") for each level of nesting.

Markdown

...

Answer Prediction

Prompting: x' = "I love this movie. Overall it was [z]"

⇩

Predicting: x' = "I love this movie. Overall it was fantastic"

- Given a prompt, predict the answer
- Use any inference algorithms, as in generation class

Post-processing

- Based on the answer, select the actual output
- For instance:
  - Taking the output as-is
  - Formatting the output for easy visualization
  - Selecting only parts of the output that you want to use
  - Mapping the outputs to other actions

Output Formatting

Markdown Rendering

You
Please write a table with the last 5 presidents and their birthdates.

ChatGPT
Here is a table with the last 5 U.S. Presidents and their birthdates:

President	Birthdate
Joe Biden	November 20, 1942
Donald Trump	June 14, 1946
Barack Obama	August 4, 1961
George W. Bush	July 6, 1946
Bill Clinton	August 19, 1946

Code

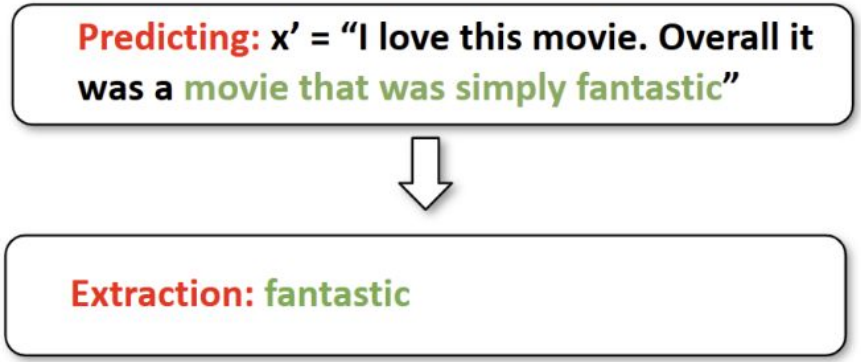
You
Output that in JSON format.

ChatGPT
Here is the information in JSON format:

```
[
  {"President":"Joe Biden","Birthdate":"November 20, 1942"},
  {"President":"Donald Trump","Birthdate":"June 14, 1946"},
  {"President":"Barack Obama","Birthdate":"August 4, 1961"},
  {"President":"George W. Bush","Birthdate":"July 6, 1946"},
  {"President":"Bill Clinton","Birthdate":"August 19, 1946"}
]
``` &#8203;`` [oaicite:0] ``&#8203;
```

- For user-facing applications, format in a pretty way
```

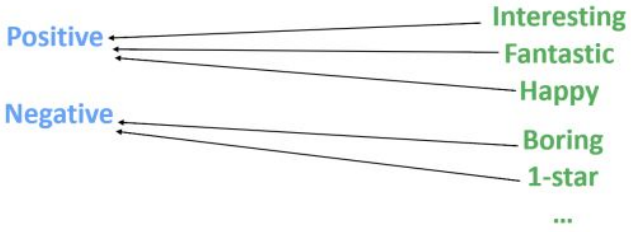
Output Selection



- From a longer response, select the information indicative of an answer
- Various methods for extraction Classification:
  - identify keywords
  - Regression/numerical problems: identify numbers
  - Code: pull out code snippets in triple-backticks

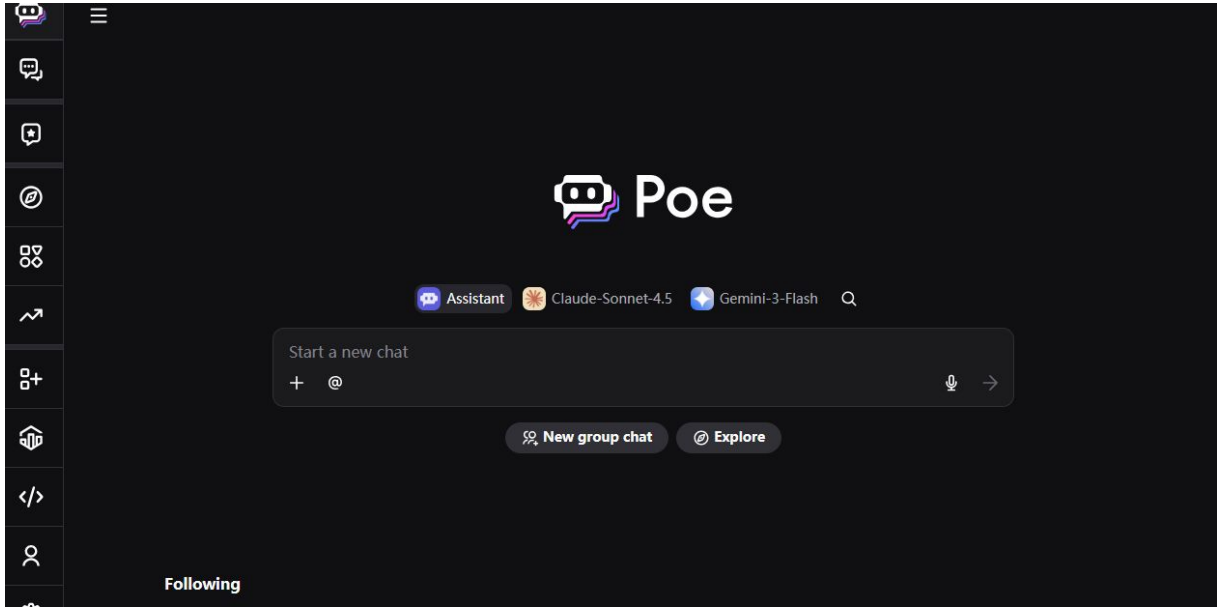
Output Mapping

Given an answer, map it into a class label or continuous value



Often map many extracted words onto a single class

Prompting playground



https://poe.com/

Few-shot Prompting

Instruction | Please classify movie reviews as 'positive' or 'negative'.

Examples

```
Input: I really don't like this movie.
Output: negative

Input: This movie is great!
Output: positive
```

- Provide a few examples of the task together with the instruction

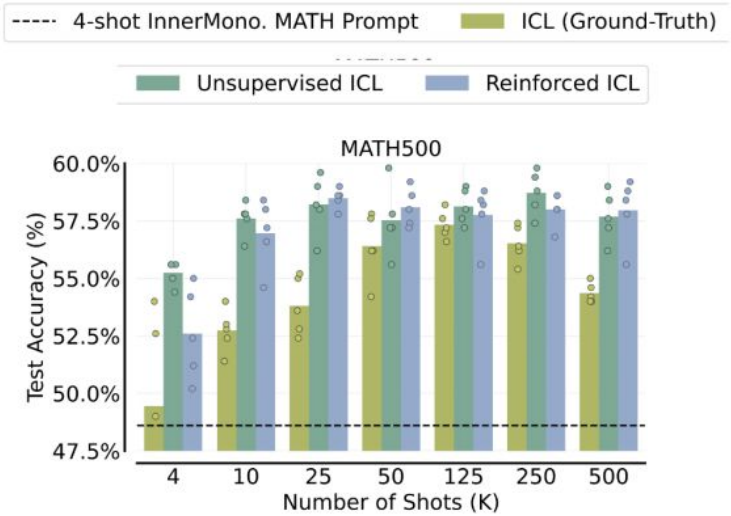
Language Models are Few-Shot Learners. NeurIPS 2020

Few-shot Prompting with Chat Prompts

```
messages=[
  {
    "role": "system",
    "content": "You are an assistant that translates corporate jargon into plain English."
  },
  {
    "role": "system",
    "name":"example_user",
    "content": "New synergies will help drive top-line growth."
  },
  {
    "role": "system",
    "name": "example_assistant",
    "content": "Things working well together will increase revenue."
  },
  ...,
  {
    "role": "user",
    "content": "This late pivot means we don't have time to boil the ocean for the client deliverable."
  },
]
```

- One approach: add “role”: “system” and a “name”: “example assistant” etc.

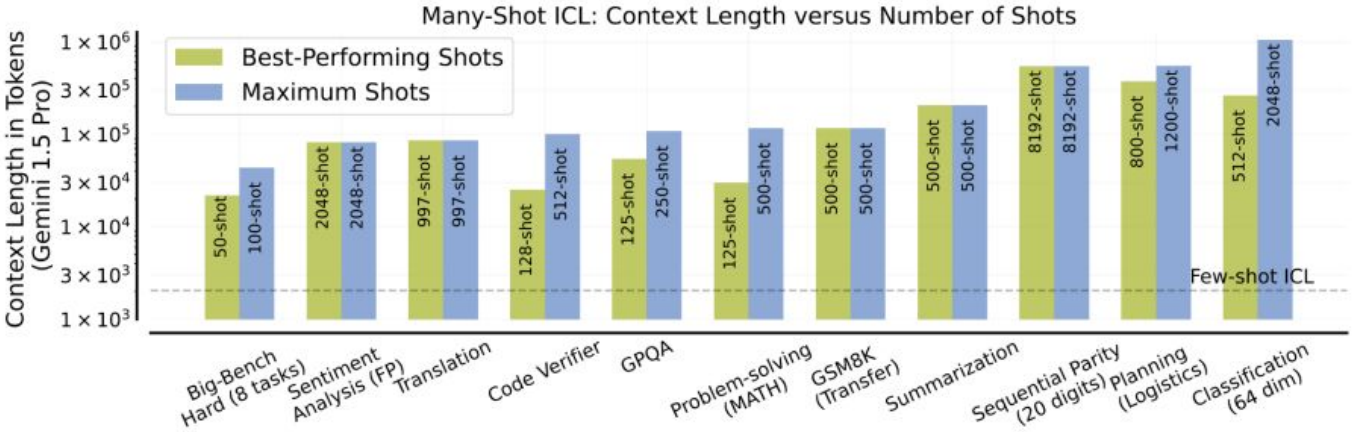
In-context learning phenomena



- Sometimes only giving the inputs works better than giving the (inputs, outputs)! ICL is more like task retrieval.

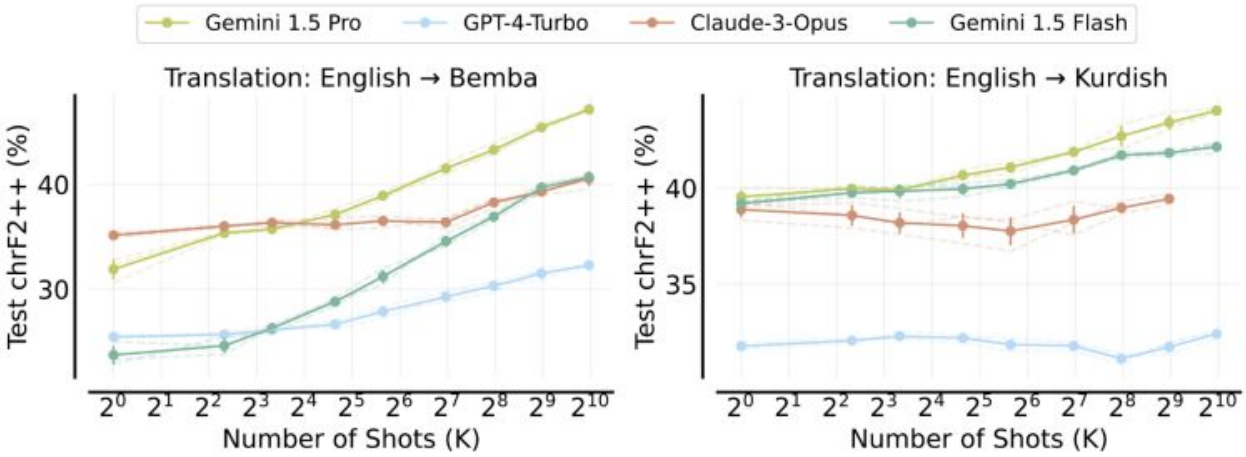
Many-Shot In-Context Learning. NeurIPS 2024

In-context learning phenomena



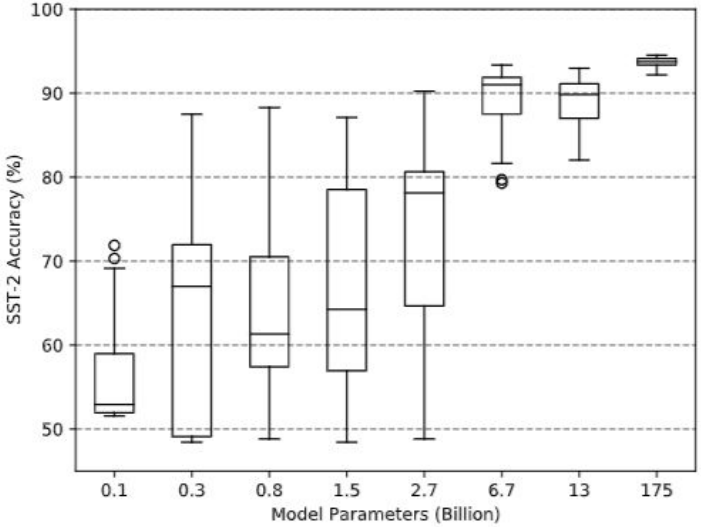
- Sometimes performance can decrease with too many examples

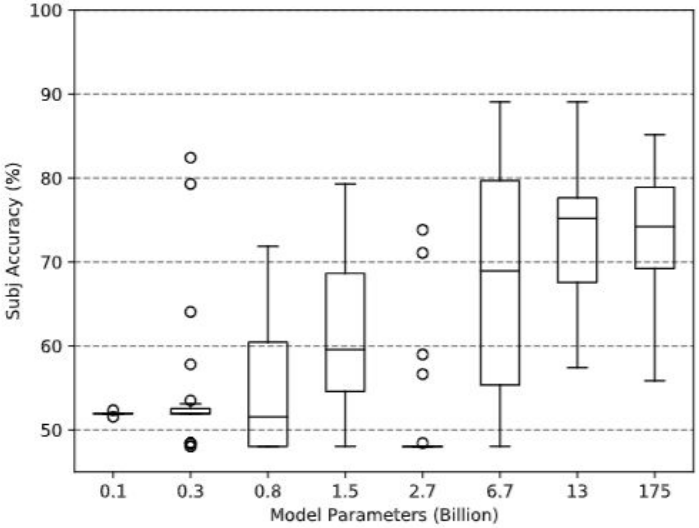
In-context learning phenomena



- Ability to leverage many examples varies by model

LMs are Sensitive to Small Changes in In-context Examples

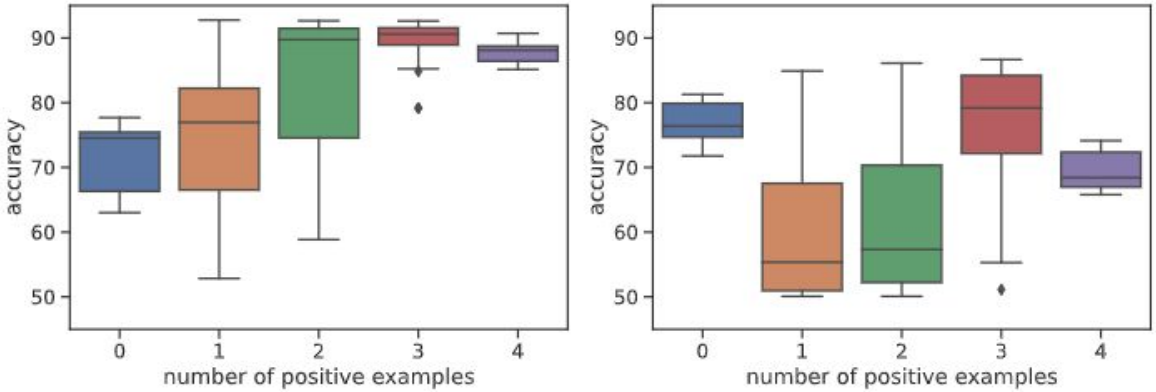




Example Orderding: some permutations have comparable performance (over 85%) to supervised training for sentiment classification, while others perform close torandom (around 50%)

Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity. ACL 2022

LMs are Sensitive to Small Changes in In-context Examples

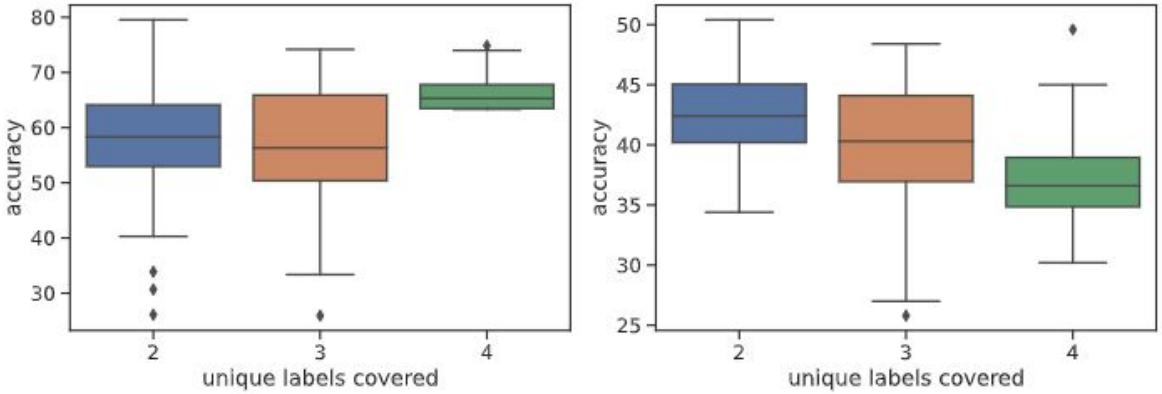


(a) Amazon (b) SST-2

Figure 3: Accuracies of Amazon and SST-2 with varying **label balance** (number of positive examples in demonstration), across 100 total random samples of 4 demonstration examples.

Active Example Selection for In-Context Learning. EMNLP 2022

LMs are Sensitive to Small Changes in In-context Examples



(a) AGNews (b) TREC

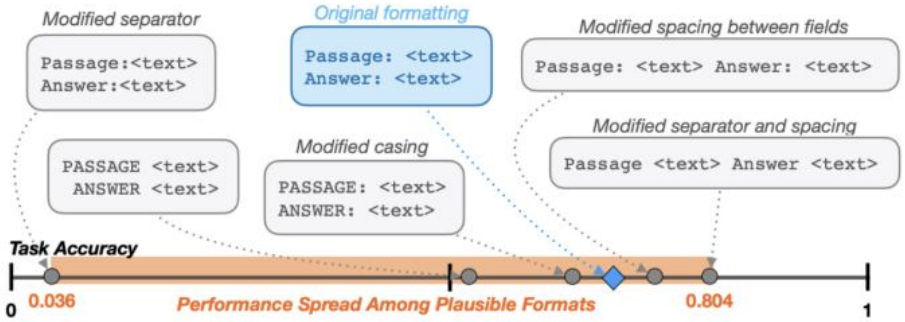
Figure 4: Accuracies of AGNews and TREC with varying **label coverage** (number of unique labels covered in demonstration), across 100 total random samples of 4 demonstration examples. Demonstration set that only covers 1 label is very unlikely and does not appear in our experiments.

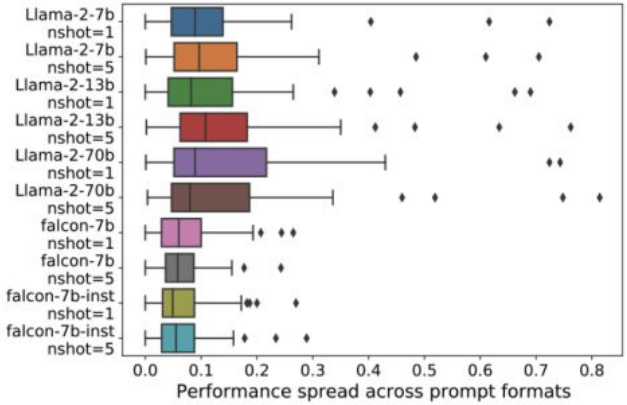
Active Example Selection for In-Context Learning. EMNLP 2022

Design of Prompts

- Manual
  - Configure a manual template based on the characteristics of the task
  - Configure prompts based on intuition about a task (later)
- Automated search
  - Search in discrete space
  - Search in continuous space

Manual Engineering: Format





- Make sure that the format matches that of a trained model
- This can have a large effect on models

Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. ICLR 2024

Manual Engineering: Instructions

- Instructions should be clear, concise and easy to understand
- Good examples: https://www.promptingguide.ai/introduction/tips
- Less Precise: Explain the concept prompt engineering. Keep the explanation short, only a few sentences, and don't be too descriptive.
- More Precise: Use 2-3 sentences to explain the concept of prompt engineering to a high school student.

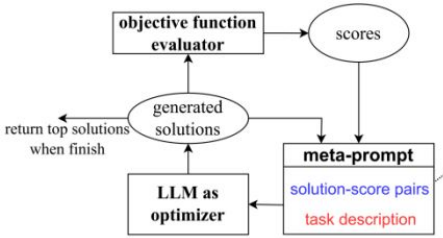
Methods for Automatic Prompt Engineering

- Using language models
- Prompt tuning
- Prefix tuning

Using language models

- Use language models to propose candidate prompts
- Evaluate the downstream task performance of each candidate
- Iterate

Example: using language models



I have some texts along with their corresponding scores. The texts are arranged in ascending order based on their scores, where higher scores indicate better quality.

text:
Let's figure it out!
score:
61

text:
Let's solve the problem.
score:
63

(… more instructions and scores …)

The following exemplars show how to apply your text: you replace <INS> in each input with your text, then read the input and give an output. We say your output is wrong if your output is different from the given output, and we say your output is correct if they are the same.

input:
Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?
A: <INS>
output:
140

(… more exemplars …)

Write your new text that is different from the old ones and has a score as high as possible. Write the text in square brackets.

- Blue: instructions.
- Find the optimal description.

Large Language Models as Optimizers. ICLR 2024

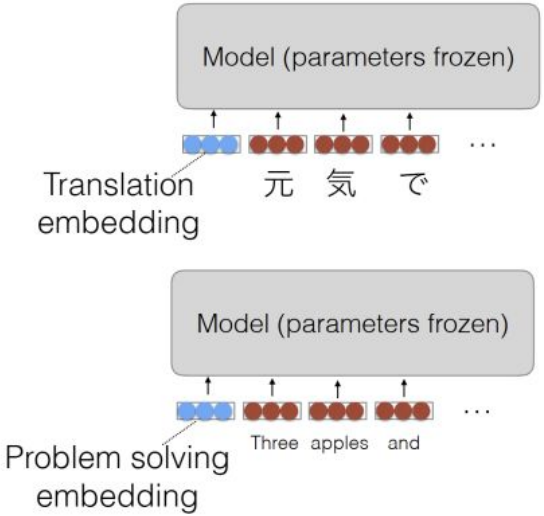
Example: using language models

Scorer	Optimizer / Source	Instruction position	Top instruction	Acc
Baselines				
PaLM 2-L	(Kojima et al., 2022)	A_begin	Let's think step by step.	71.8
PaLM 2-L	(Zhou et al., 2022b)	A_begin	Let's work this out in a step by step way to be sure we have the right answer.	58.8
PaLM 2-L		A_begin	Let's solve the problem.	60.8
PaLM 2-L		A_begin	(empty string)	34.0
Ours				
PaLM 2-L	PaLM 2-L-IT	A_begin	Take a deep breath and work on this problem step-by-step.	**80.2**
PaLM 2-L	PaLM 2-L	A_begin	Break this down.	79.9
PaLM 2-L	gpt-3.5-turbo	A_begin	A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem.	78.5
PaLM 2-L	gpt-4	A_begin	Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer.	74.5

- Task: solve grade-school math word problems (GSM8k)

Large Language Models as Optimizers. ICLR 2024

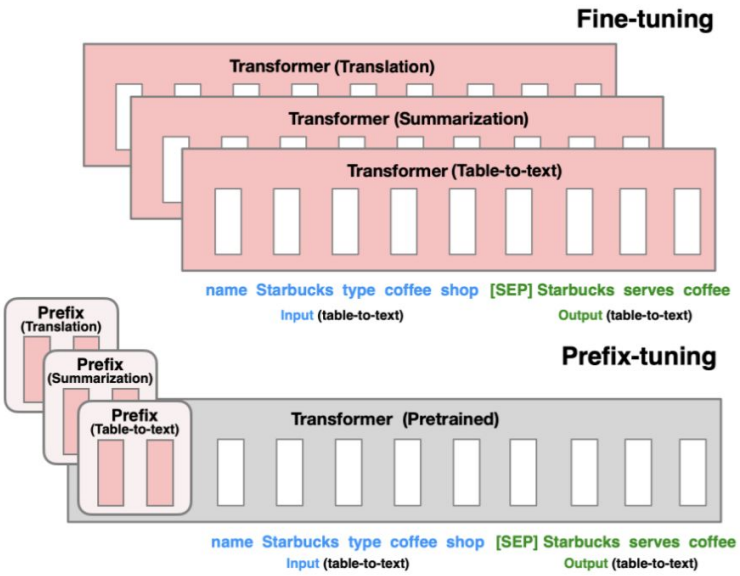
Prompt Tuning



- Optimize the embeddings of a prompt, instead of the words.

The Power of Scale for Parameter-Efficient Prompt Tuning. EMNLP 2021

Prefix Tuning



- Prefix tuning: optimize vectors for attention keys/values

Prefix-Tuning: Optimizing Continuous Prompts for Generation. ACL 2021

Chain-of-Thought Prompting

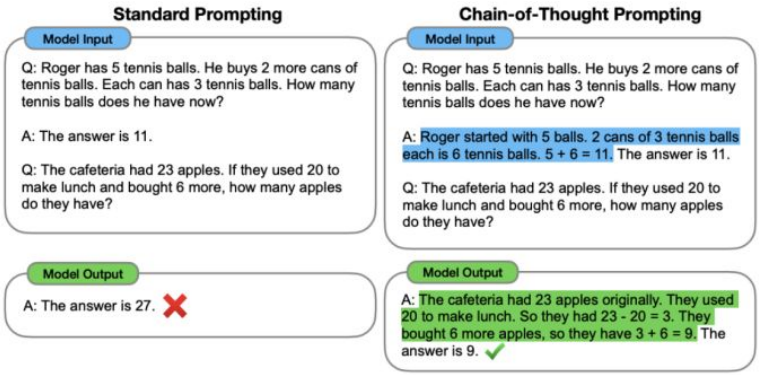


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

- Get the model to explain its reasoning before making an answer
- Provides the model with adaptive computation time

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022

Chain-of-thought without examples

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Kojima et al 2022
(proprietary models PaLM and Instruct-GPT)

Models may output reasoning chains without examples

- Chain-of-thought style data in pretraining/fine-tuning data
- System prompts

Large Language Models are Zero-Shot Reasoners. NeurIPS 2022

Program-aided Language Models

Program-aided Language models (this work)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 tennis balls.
`tennis_balls = 5`
2 cans of 3 tennis balls each is
`bought_balls = 2 * 3`
tennis balls. The answer is
`answer = tennis_balls + bought_balls`

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves of bread did they have left?

Model Output

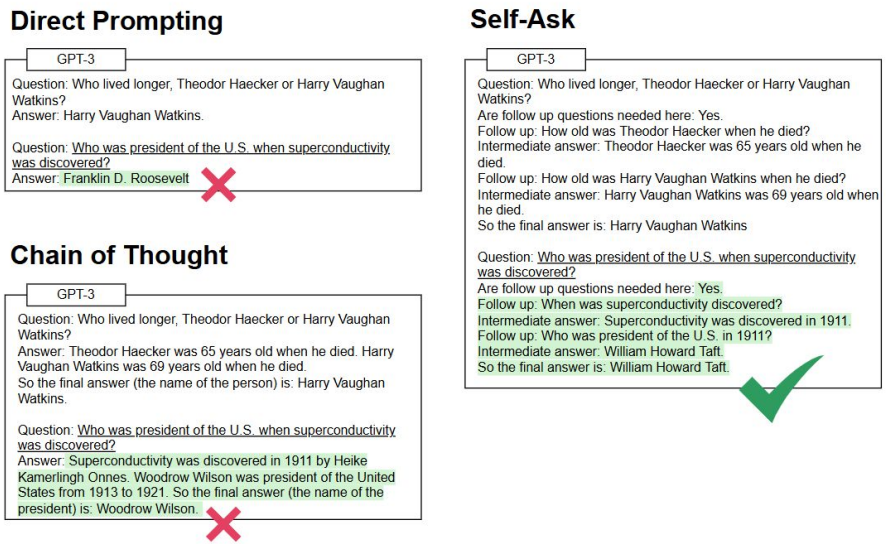
A: The bakers started with 200 loaves
`loaves_baked = 200`
They sold 93 in the morning and 39 in the afternoon
`loaves_sold_morning = 93`
`loaves_sold_afternoon = 39`
The grocery store returned 6 loaves.
`loaves_returned = 6`
The answer is
`answer = loaves_baked - loaves_sold_morning - loaves_sold_afternoon + loaves_returned`

```
>>> print(answer)
74
```

- Using a program to generate outputs can be more precise than asking the LM to do so
- Especially useful for numeric questions

PAL: Program-aided Language Models. ICML 2023

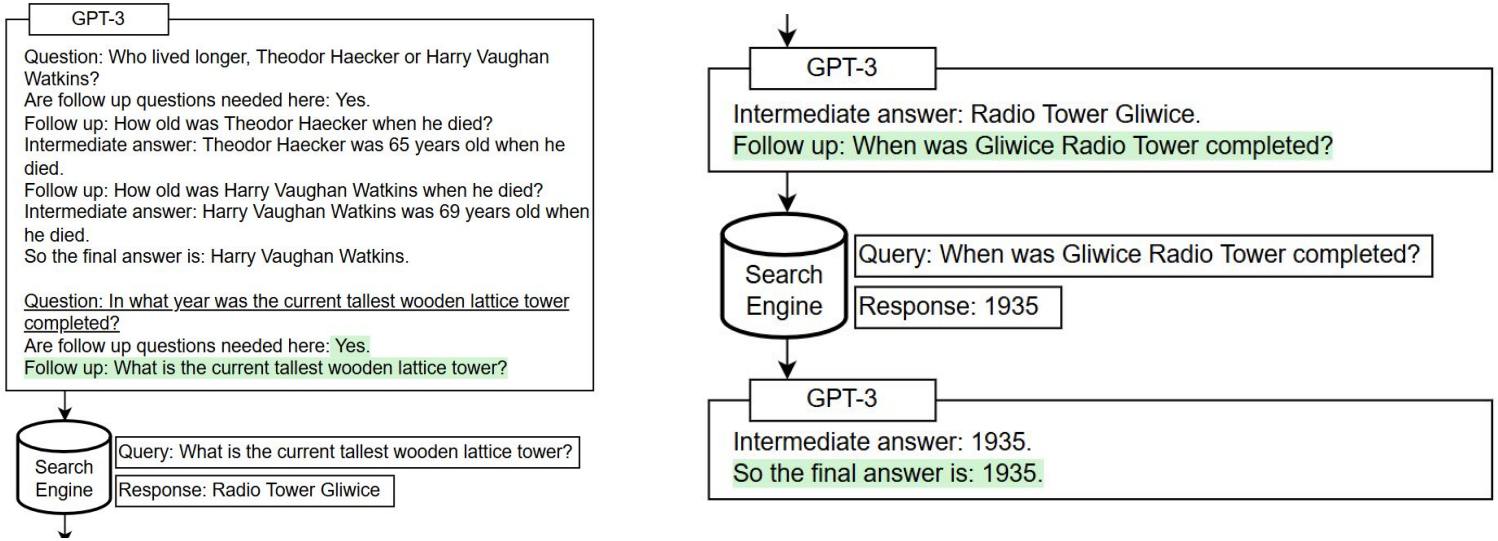
Self-Ask: Problem decomposition



- Decompose problems into multiple steps: 1. Ask sub-question; 2. Query search engine; 3. Repeat

Measuring and Narrowing the Compositionality Gap in Language Models. Findings of EMNLP 2023

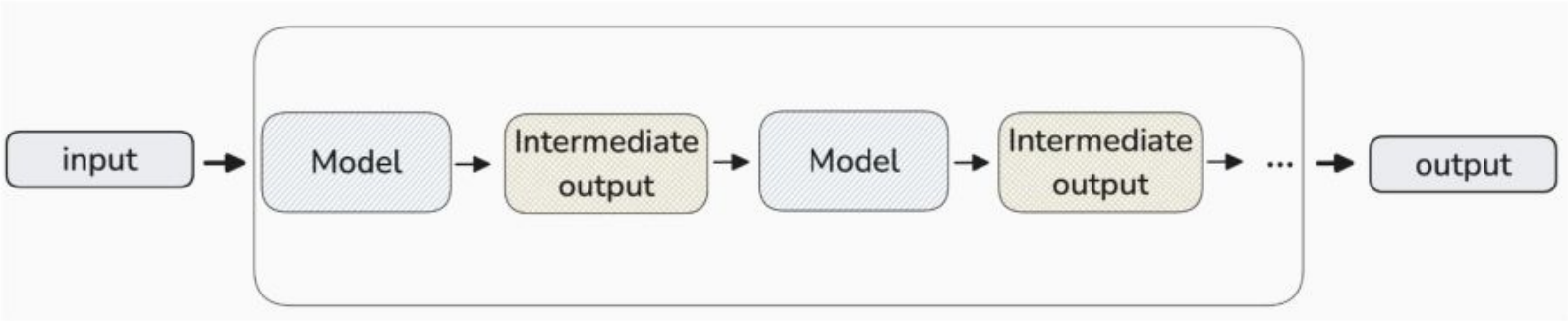
Self-Ask: Problem decomposition



- Decompose problems into multiple steps: 1. Ask sub-question; 2. Query search engine; 3. Repeat

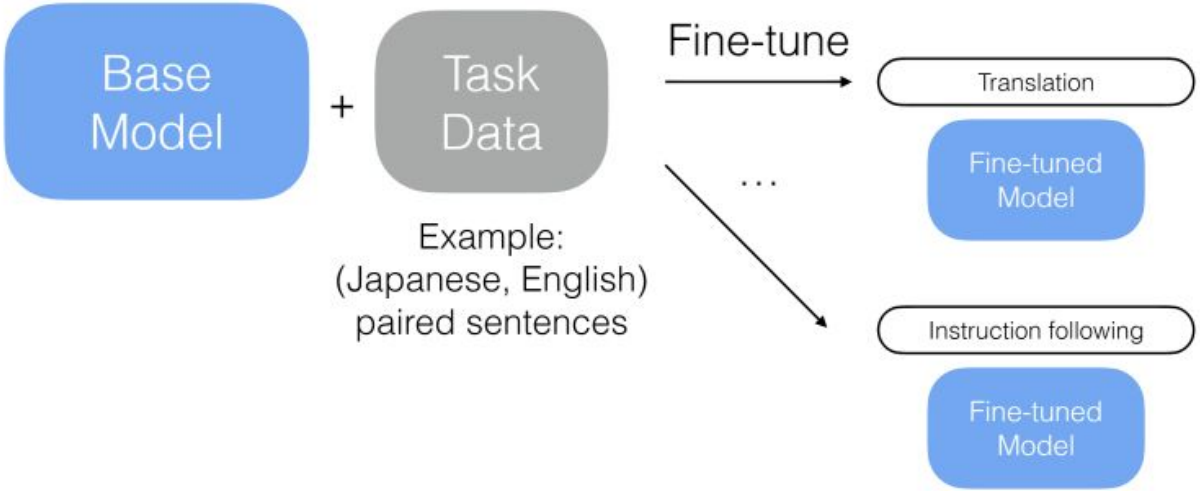
Measuring and Narrowing the Compositionality Gap in Language Models. Findings of EMNLP 2023

Prompt chains



- More generally, “chain” together multiple calls to prompted models (and/or external functions)

Adaptation: fine-tune



Standard LM fine-tuning

- Initialize a model $p_\theta(y|x)$ with a pre-trained model
- Train the model on a dataset $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$
- $$\arg\min_\theta \sum_{(x,y)\in D} \sum_t -\log p_\theta(y_t|y_{<t}, x)$$

Examples


```
{
    "abstract": "\" we have studied the leptonic decay @xmath0 , vi
    "article": "\"the leptonic decays of a charged pseudoscalar mes
    "section_names": "[sec:introduction]introduction\n[sec:detector
}
```


- X: Article (paper)
- Y: Abstract
- Training set: 300,000+ examples

Examples

```
{'problem': 'A board game spinner is divided into three parts label
 'level': 'Level 1',
 'type': 'Counting & Probability',
 'solution': 'The spinner is guaranteed to land on exactly one of t
```

- X: Problem
- Y: Solution
- Training set: 7,500 examples

Examples

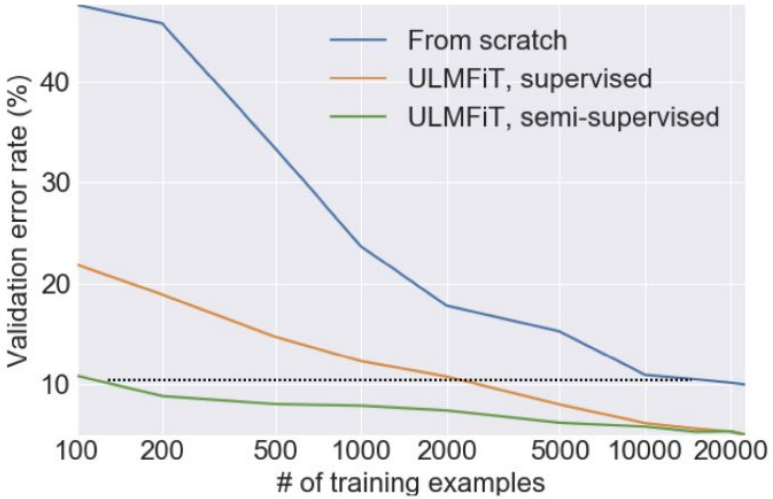
question string · *lengths*	response string · *lengths*
12↔4.07k 96.5% How does the sentence end?…	0↔1.5k 92.9% B). Rebecca was saddened by people's vile insults. Here's why: "Thicker skin" is a way to say that…
Write a sentence not in English.	Voici une phrase en français: "Le chat mange la souris." Explication pour un enfant de cinq ans: Cette phrase n'est pas en anglais, mais en français. Elle dit que le chat (l'animal avec des moustaches et qui dit "miaou") mange la souris (un petit animal avec une longue queue).

- X: Prompt
- Y: Response
- Training set: 2,910,000 examples

“No code” fine-tuning libraries

- Llama Factory,
- Axolotl

Effects of fine-tuning



- Starting from a pre-trained model is data-efficient

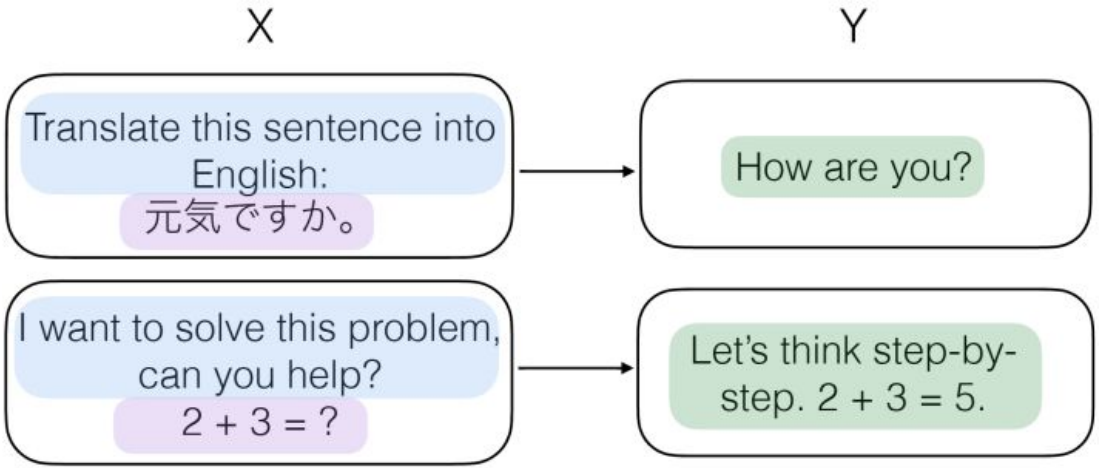
Effects of fine-tuning

- "Narrows" the distribution
- Pre-training: minimize $D_{KL}(p_{\text{data}}, p_\theta)$
- Fine-tuning: minimize $D_{KL}(p_{\text{data fine-tune}}, p_\theta; p_0)$
- Typically the pretraining data will cover a wider distribution than the fine-tuning data

Effects of fine-tuning

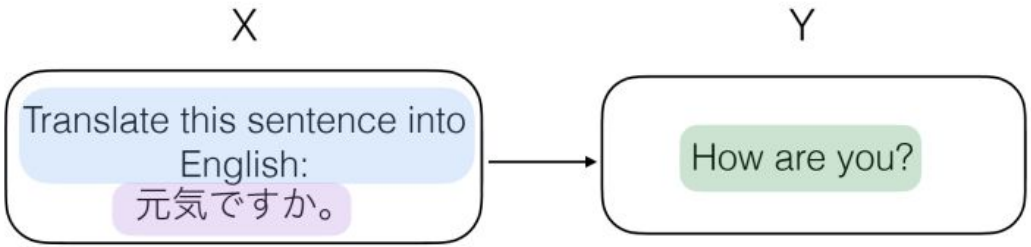
- Example symptoms:
  - Summarization model doesn’t work well on translation
  - Model trained with specific formatting requires the formatting
  - Model can’t few-shot learn well after fine-tuning

Instruction Tuning



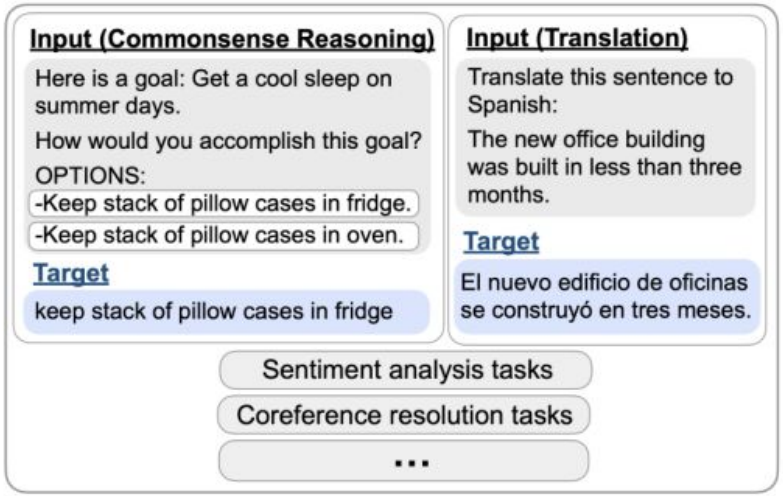
- Fine-tune a model to perform multiple tasks
- Insight: use (instruction + input, output) data

Variations



- Instructions: template, human, model-generated
- Input: dataset, human, model-generated
- Output: dataset, human, model-generated
- Domain: general, code, math, chat, . . .

Example: FLAN



- 62 NLP datasets
- Instructions: templates
- Input: from dataset
- Output: from dataset

Finetuned Language Models Are Zero-Shot Learners. ICLR 2022

Example: FLAN

Template 1

<premise>

Based on the paragraph above, can we conclude that <hypothesis>?

<options>

Template 2

<premise>

Can we infer the following?

<hypothesis>

<options>

Template 3

Read the following and determine if the hypothesis can be inferred from the premise:

Premise: <premise>

Hypothesis: <hypothesis>

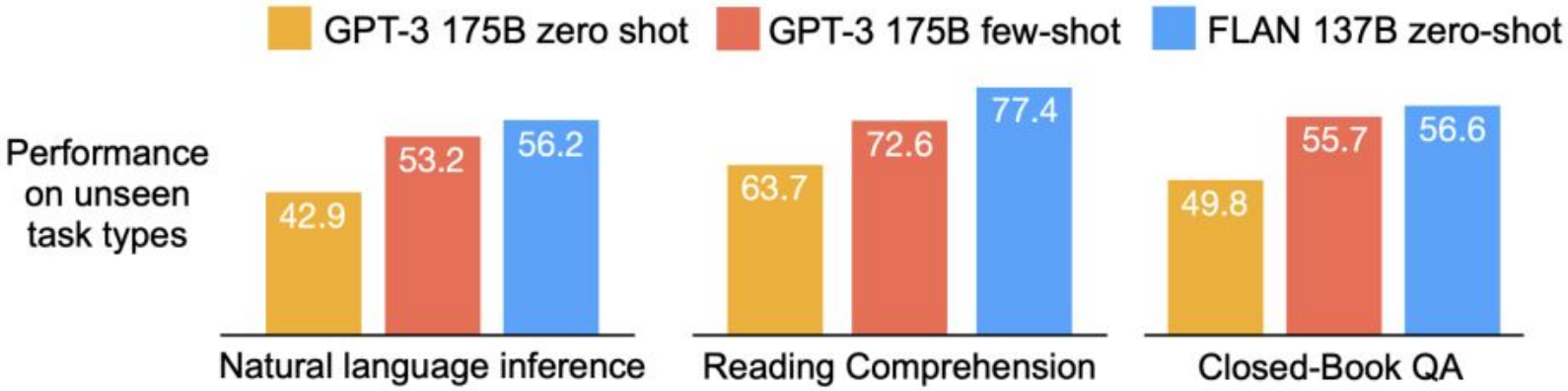
<options>

Template 4, …

- 62 NLP datasets
- Instructions: templates
- Input: from dataset
- Output: from dataset

Finetuned Language Models Are Zero-Shot Learners. ICLR 2022

Example: FLAN



- Key finding: model can generalize to unseen tasks

Finetuned Language Models Are Zero-Shot Learners. ICLR 2022

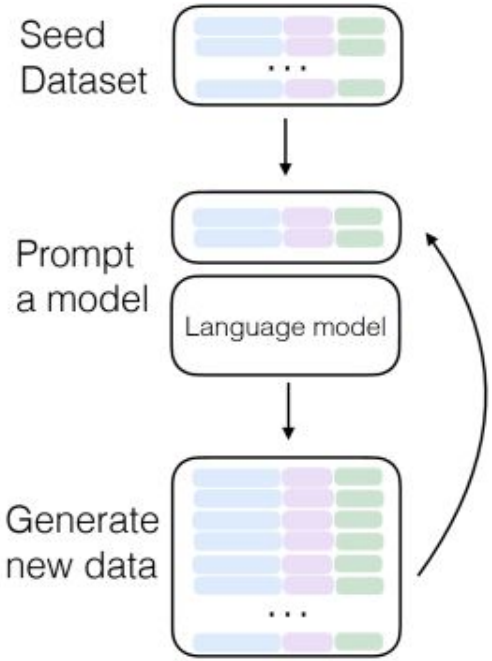
Example: SuperNaturalInstructions

- 1,600 tasks
- Instructions: crowdsourced
- Input: crowdsourced
- Output: crowdsourced

Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks. EMNLP 2022

Example: Self-Instruct

- 50,000+ instructions
- Instructions: model
- Input: model
- Output: model



Self-Instruct: Aligning Language Models with Self-Generated Instructions. ACL 2023

Example: Self-Instruct

- 50,000+ instructions
- Instructions: model
- Input: model
- Output: model

Instruction: Given an address and city, come up with the zip code.
Input:
Address: 123 Main Street, City: San Francisco
Output: 94105

Instruction: I am looking for a job and I need to fill out an application form. Can you please help me complete it?
Input:
Application Form:
Name:_____ Age:_____ Sex:_____
Phone Number:_____ Email Address:_____
Education:_____ ...
Output:
Name: John Doe Age: 25 Sex: Male
Phone Number: ...

Instruction: How to write a code for converting degrees fahrenheit to celsius.
Input: Null
Output:

```
def convert_fahrenheit_to_celsius(fahr):
    celsius = (fahr - 32) * 5 / 9
    return celsius
```

Self-Instruct: Aligning Language Models with Self-Generated Instructions. ACL 2023

Recap: instruction tuning

- Finetune on (instruction, input, output) examples
- Get data from humans or model generations
- Resulting model can perform many tasks
- Lots of followup work on instruction-tuning data for certain domains: general, code, math, chat, . . .

Chat tuning

System

User

Assistant

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>

You are a helpful AI assistant for travel tips and
recommendations<|eot_id|>
<|start_header_id|>user<|end_header_id|>

What can you help me with?<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Ultimately, format a chat as a sequence of tokens

- System prompt
- (user, assistant, user, assistant, ...)

Chat tuning: system prompts

System Instruction: You are an AI assistant. User will you give you a task. Your goal is to complete the task as faithfully as you can. While performing the task think step-by-step and justify your steps.

User Instruction: Use the given data to calculate the median.

Input: [7, 3, 8, 2, 10]

Output: To calculate the median, I will follow these steps:
Step 1: Arrange the data in ascending order [2, 3, 7, 8, 10]
Step 2: Determine if the data set has an odd or even number of values
The data set has 5 values, which is an odd number.
Step 3: Find the middle value

Since the data set has an odd number of values, the median is the middle value in the ordered list. The middle value in the list [2, 3, 7, 8, 10] is 7.

System Instruction: You are an AI assistant. Provide a detailed answer so user don't need to search outside to understand the answer.

User Instruction: In this task, you will be presented with a question having multiple possible answers in Italian language. And you should choose a most suitable option out of "A", "B", "C", "D", and "E" based on your commonsense knowledge.

Input: Solve this question: Dove non riusciresti a vedere la luce?
Options: A scrivania B frigorifero C sole D universo E attico

Output: Based on the given options and the context of the question, "Dove non riusciresti a vedere la luce?" (Where would you not be able to see light?), the most suitable answer would be: B frigorifero (refrigerator).

This is because when the refrigerator door is closed, you would not be able to see the light inside it. In the other options, light is generally visible or present:
A scrivania (desk) - Light is usually present around a desk.
C sole (sun) - The sun is the primary source of light in our solar system.
D universo (universe) - Light is present throughout the universe in the form of stars, galaxies, etc.
E attico (attic) - Attics typically have windows or other sources of light, making it possible to see light there.

- Example: OpenOrca
- System messages: 16 hand-written
- Outputs: from GPT-4

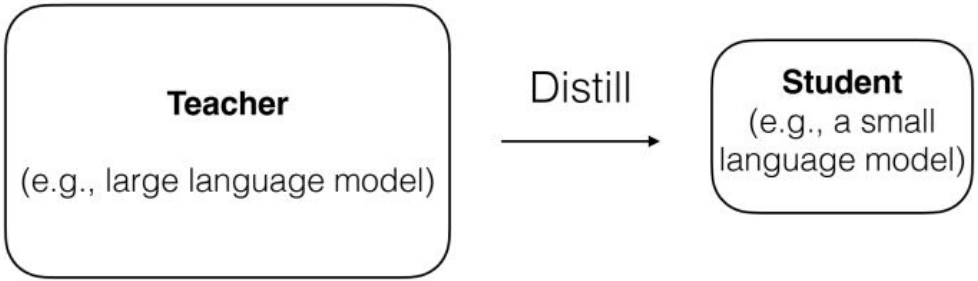
Chat tuning: system prompts

> You are a helpful and cheerful AI Chatbot that acts as a meal plan assistant for busy families. The family consists of 2 adults, 3 teenagers, and 2 preschoolers. Plan two or three days at a time and use leftovers or extra ingredients for the second day's plan. The user will let you know if they want two or three days. If they don't, assume three days. Each plan should include breakfast, lunch, snack, and dinner. Ask the user if they approve of the plan or need adjustments. After they approve provide a grocery list with family size in mind. Always keep family preferences in mind and if there's something that they don't like provide a substitution. If the user is not feeling inspired then ask them what's the one place they wish they could visit on vacation this week and then suggest meals based on that location's culture. Weekend meals can be more complex. Weekday meals should be quick and easy. For breakfast and lunch, easy food like cereal, English muffins with pre-cooked bacon, and other quick easy foods are preferred. The family is busy. Be sure to ask if they have essentials and favorites on hand like coffee or energy drinks so they don't forget to buy it. Remember to be budget-conscious unless it's a special occasion.

Example: Llama 3

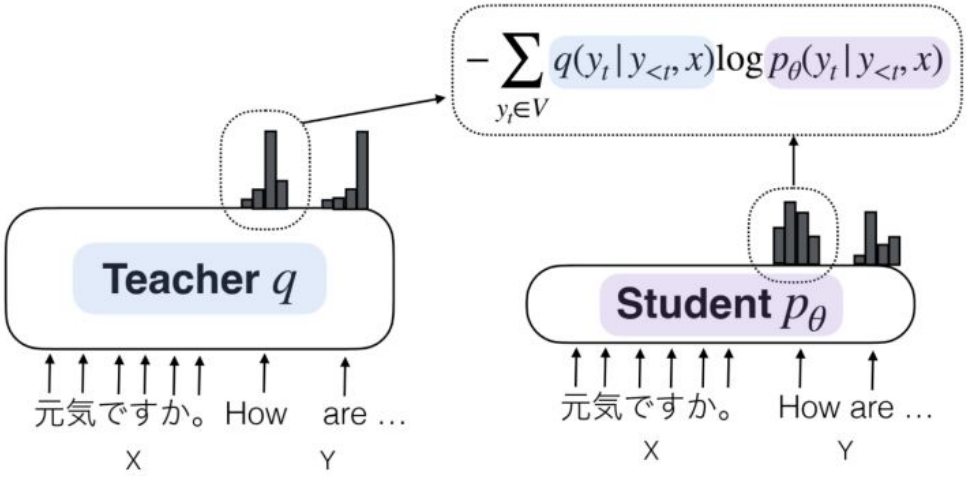
- Human-written
- Other mentions of system prompts, no details

Knowledge distillation



- Several methods we discussed use a good model (e.g., GPT-4) to generate data for another model
- Instance of knowledge distillation

Token-level knowledge distillation



- Train student to mimic teacher's token distributions
- Distillation loss (cross entropy)

Distilling the Knowledge in a Neural Network. *NIPS 2014 Deep Learning Workshop*

Token-level knowledge distillation

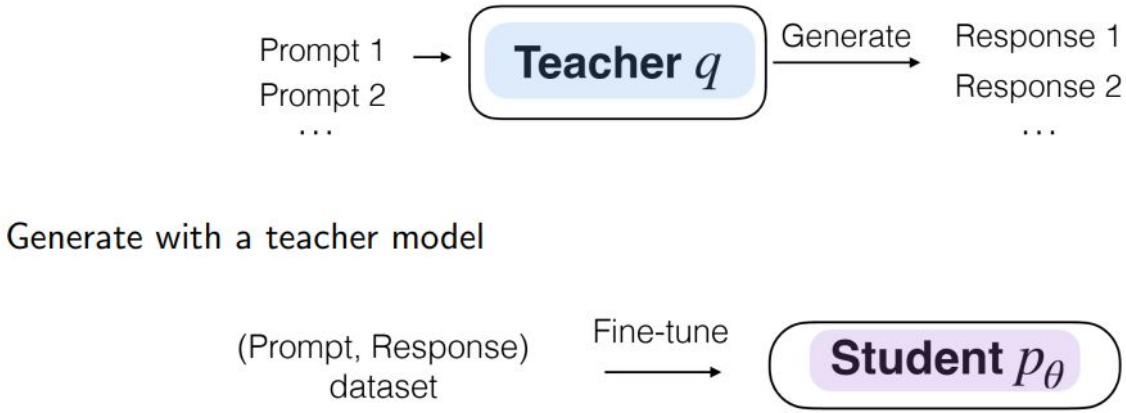
- Minimizes KL between teacher and student:

$$\min_{\theta} KL\left(q(y\,|\,x)\,\|\,p_\theta(y\,|\,x)\right)$$

$$\equiv \min_{\theta} \mathbb{E}_{y\sim q(y|x)}\left[\sum_{t}\sum_{y_t\in V} -\underbrace{q(y_t\,|\,y_{<t},x)}_{\text{“Soft labels”}}\log p_\theta(y_t\,|\,y_{<t},x)\right]$$

Distilling the Knowledge in a Neural Network. NIPS 2014 Deep Learning Workshop

Sequence-level knowledge distillation



- Student model fine-tunes on the generated data

Sequence-Level Knowledge Distillation. EMNLP 2016

Sequence-level knowledge distillation

- Also minimizes KL between teacher and student:

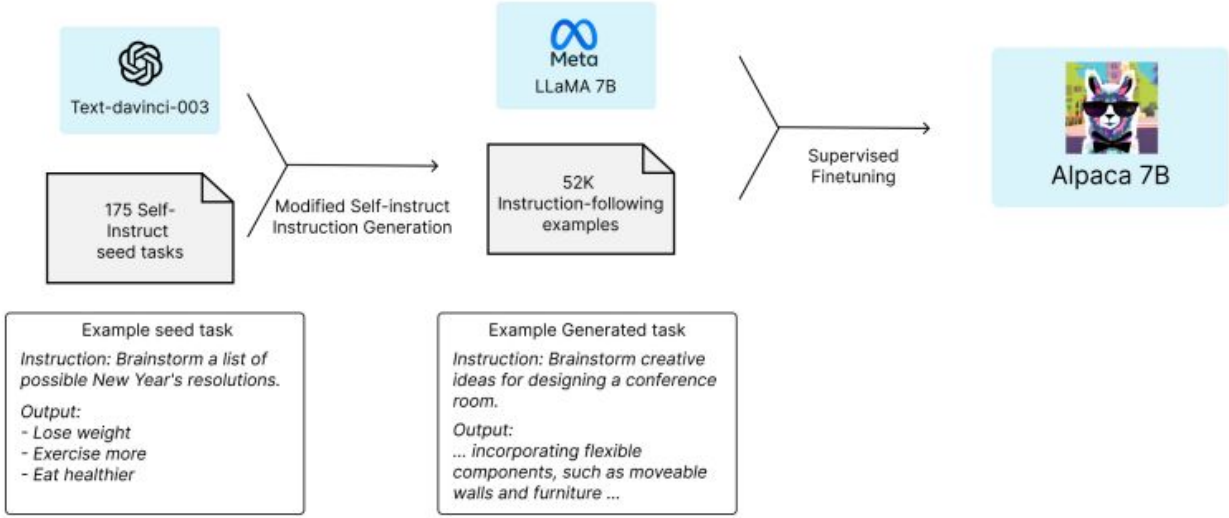
$$\min_{\theta} KL\left(q(y\,|\,x)\,\|\,p_\theta(y\,|\,x)\right)$$

$$\equiv \min_{\theta} \mathbb{E}_{y\sim q(y|x)}\left[-\log p_\theta(y\,|\,x)\right]$$

Teacher generations

Sequence-Level Knowledge Distillation. EMNLP 2016

Sequence-level knowledge distillation



Sequence-level knowledge distillation

Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* ChatGPT Quality

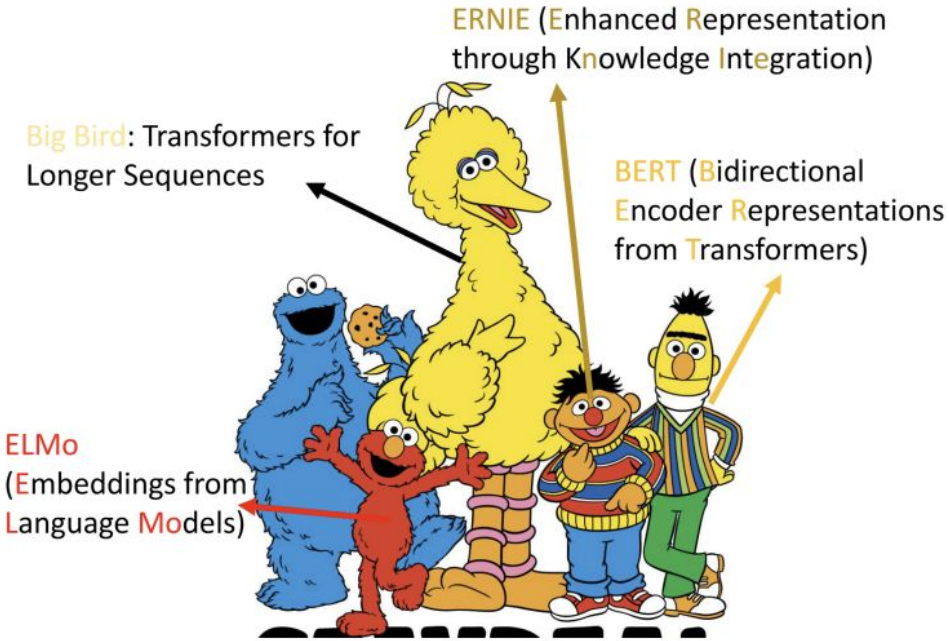
by: The Vicuna Team, Mar 30, 2023

We introduce Vicuna-13B, an open-source chatbot trained by fine-tuning LLaMA on user-shared conversations collected from ShareGPT. Preliminary evaluation using GPT-4 as a judge shows Vicuna-13B achieves more than 90%* quality of OpenAI ChatGPT and Google Bard while outperforming other models like LLaMA and Stanford Alpaca in more than 90%* of cases. The cost of training Vicuna-13B is around $300. The code and weights, along with an online demo, are publicly available for non-commercial use.

Vicuna (generated by stable diffusion 2.1)

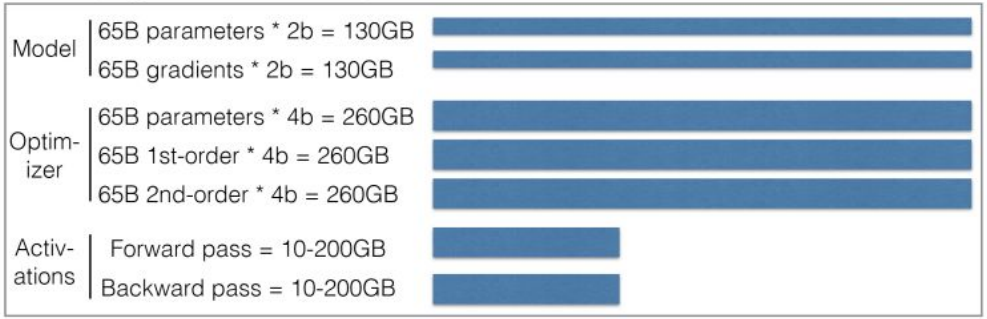
*According to a fun and non-scientific evaluation with GPT-4. Further rigorous evaluation is needed.

PLM: Sesame Street



LLM: Llama Family

Full Fine-tuning



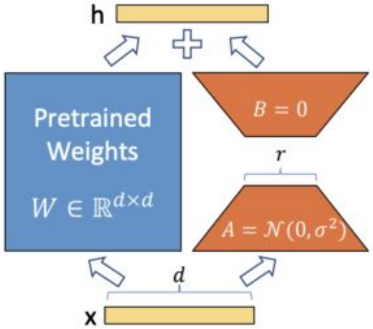
1000-1400GB of GPU memory!

- Simply continue training the LM on the output
- Issue: depending on optimizer, optimization method, can take lots of memory!
- Example: Training 65B parameter model with 16-bit mixed precision (Rajbhandari et al. 2019)

An Aside: GPU Specs

GPU	Memory	Cost (2/2024)	(Cloud) Machines
T40 / K80	24GB	$150	Google Colab, AWS p2.*
V100	32GB	$2,500	Google Colab
A100	40GB or 80GB	$8,000/$16,000	Google Colab, AWS p3.*
H100	80GB	$44,000	AWS p4.*
6000 Ada, L40	48GB	$8000	N/A
Mac M*	Same as CPU	$2000	N/A

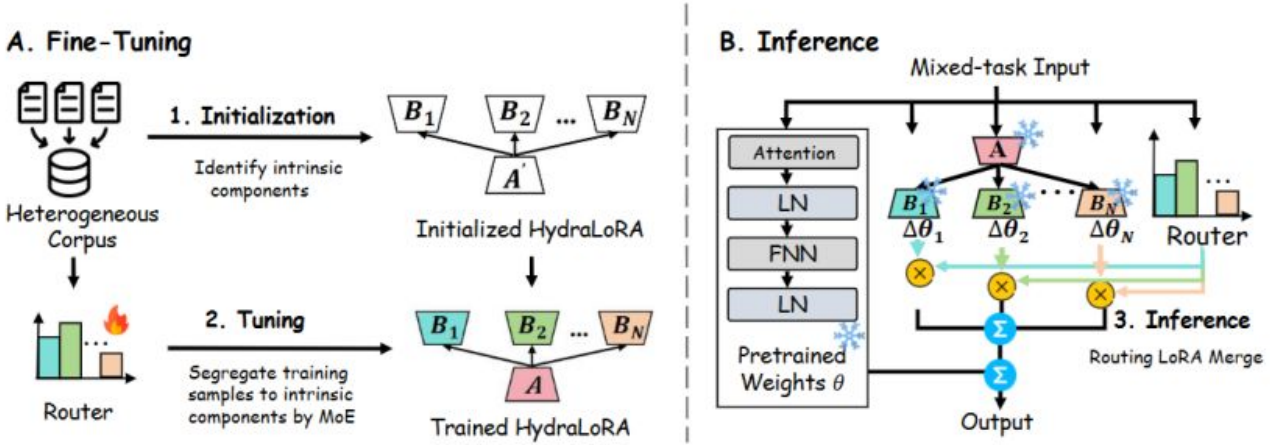
Low-Rank Adaptation (LoRA)



- Freeze pre-trained weights, train low-rank approximation of difference from pre-trained weights
- $\underline{W'} = \underline{W} + \underline{A} \quad \underline{B}$
- $\mathbb{R}^{d \times d} \quad \mathbb{R}^{d \times r} \quad \mathbb{R}^{r \times d}$
- After training, just add in to pre-trained weights!

LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022

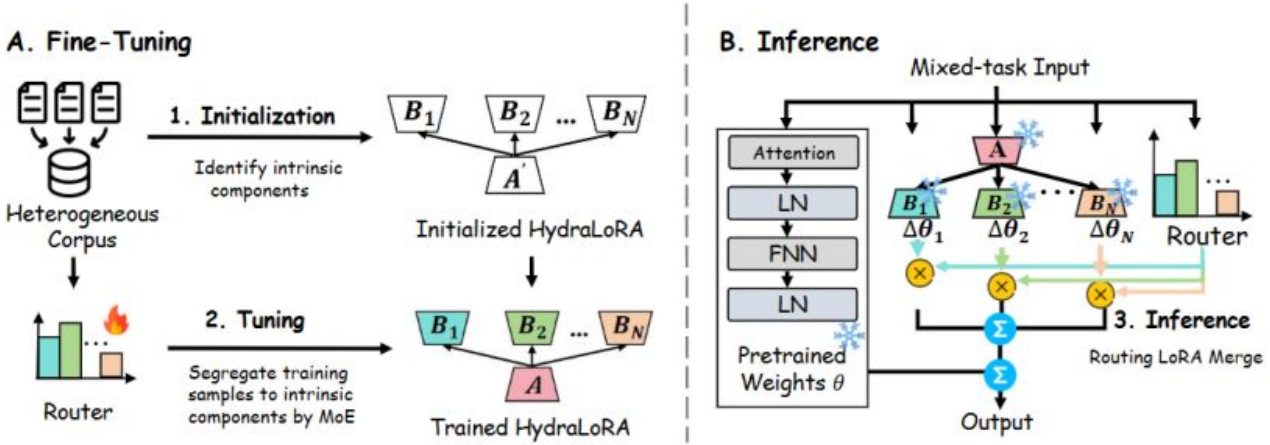
HydraLora



- Using a single LoRA for an entire domain can lead to performance degradation due to task interference, especially in datasets with diverse sub-domains or tasks.

An Asymmetric LoRA Architecture for Efficient Fine-Tuning. NeurIPS (Oral) 2024

HydraLora



- HydraLoRA introduces a **shared A matrix (capturing commonalities across tasks)** and **multiple B matrices (task-specific adaptations)**, reducing parameter redundancy while enhancing task-specific learning.

An Asymmetric LoRA Architecture for Efficient Fine-Tuning. NeurIPS (Oral) 2024

Unified View of Parameter-Efficient Fine-Tuning (PEFT)

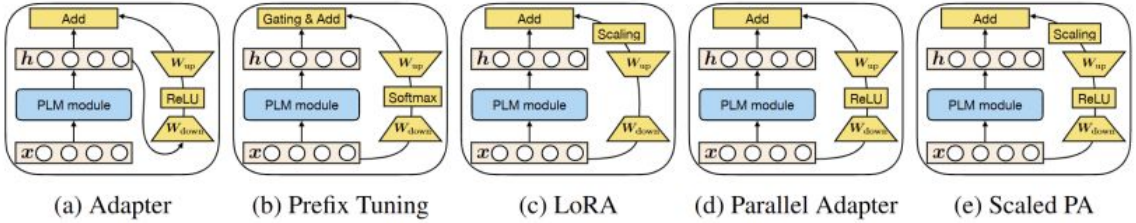


Figure 3: Graphical illustration of existing methods and the proposed variants. "PLM module" represents a certain sublayer of the PLM (e.g. attention or FFN) that is frozen. "Scaled PA" denotes scaled parallel adapter. We do not include multi-head parallel adapter here to save space.

- Reveals connections among state-of-the-art PEFT (including adapter tuning, prefix tuning, and LoRA);
- Frames these methods as modifications to specific hidden states of frozen PLMs, defining key design dimensions like the functional form of modification, insertion form, modified representation, and composition function.

Towards a Unified View of Parameter-Efficient Transfer Learning. ICLR (Spotlight) 2024

Recap

- Fine-tuning basics: Adjust a model's parameters using data
- Instruction tuning: Format data so that a model learns to do multiple tasks
- Knowledge distillation: Data can come from various teachers (human, model)
- Efficient fine-tuning: Only update some of the parameters

Recap

- So far:
  - Pretraining
  - Prompting
  - Finetuning
- Will cover:
  - Reinforcement Learning with Human Feedback
  - Multimodal Foundation Mode

Problem 1: task mismatch

We typically want a model to perform well at tasks

Language model		Task criterion
$p(\text{probable response} \mid \text{prompt})$	$\approx$	Helpful response Non-offensive response
$p(\text{probable solution} \mid \text{problem})$	$\approx$	Correct solution Code that passes test cases

Problem 2: data mismatch

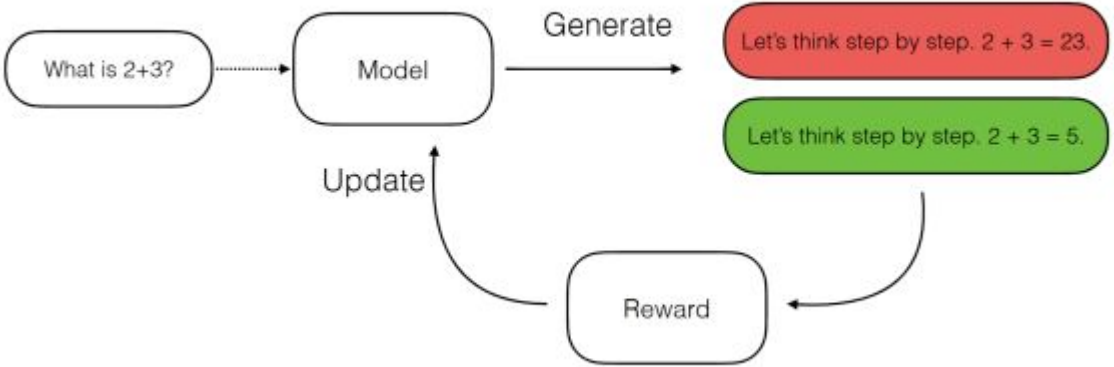
- Data often contains outputs we don’t want
  - Toxic / offensive comments from Reddit
  - Buggy code
- We don’t have much task-specific data
  - Chains of thought while solving problems
  - Helpful responses to all prompts

Problem 3: exposure bias

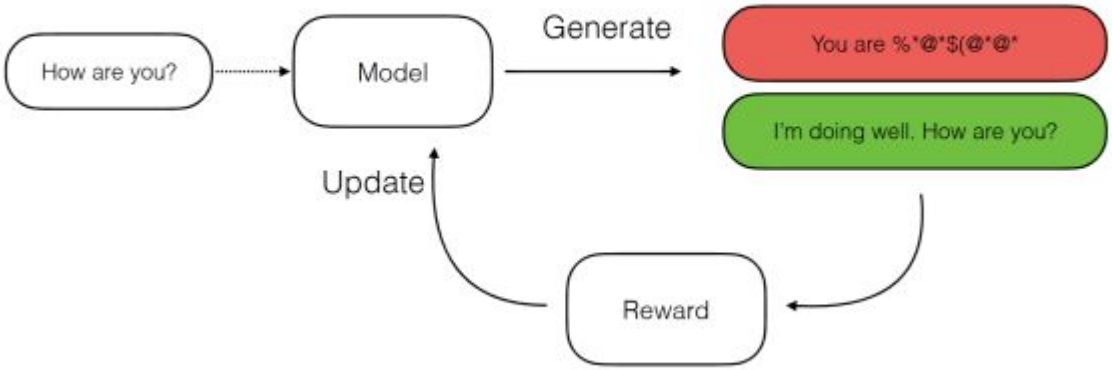
The model is not exposed to mistakes during training, and cannot deal with them at test-time

- E.g., make a mistake while solving a problem
- E.g., click the wrong page while buying something online

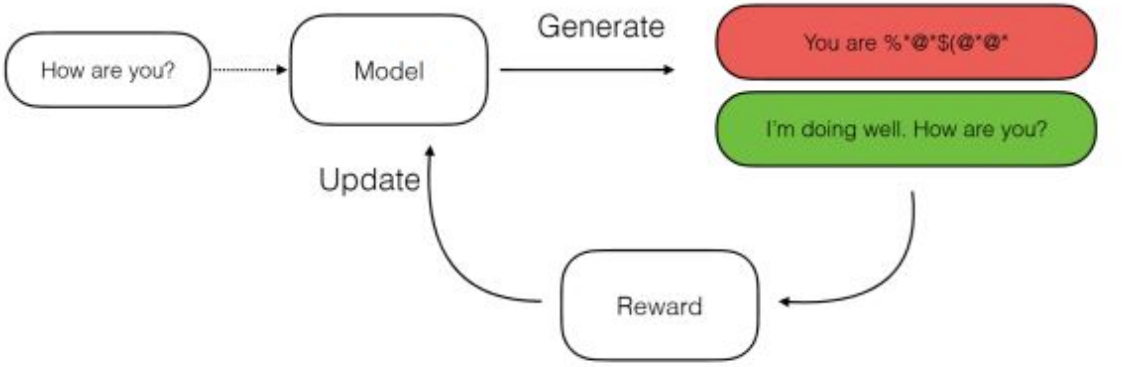
Reinforcement Learning



Reinforcement Learning

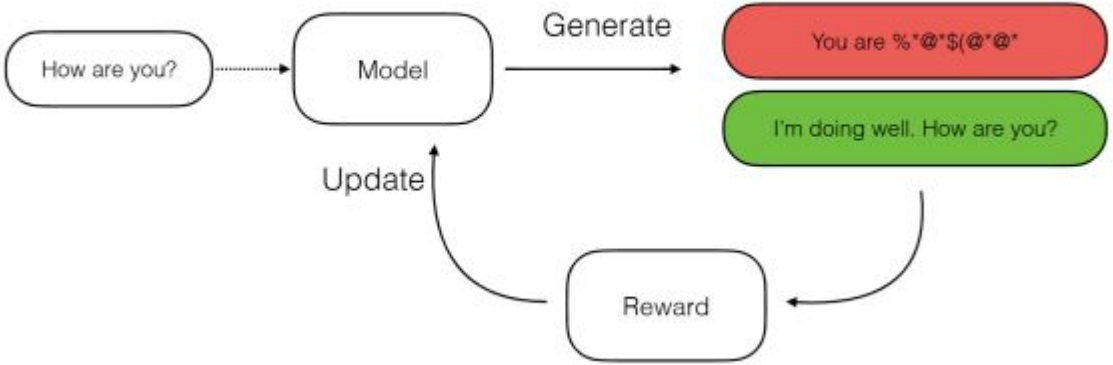


Reinforcement Learning



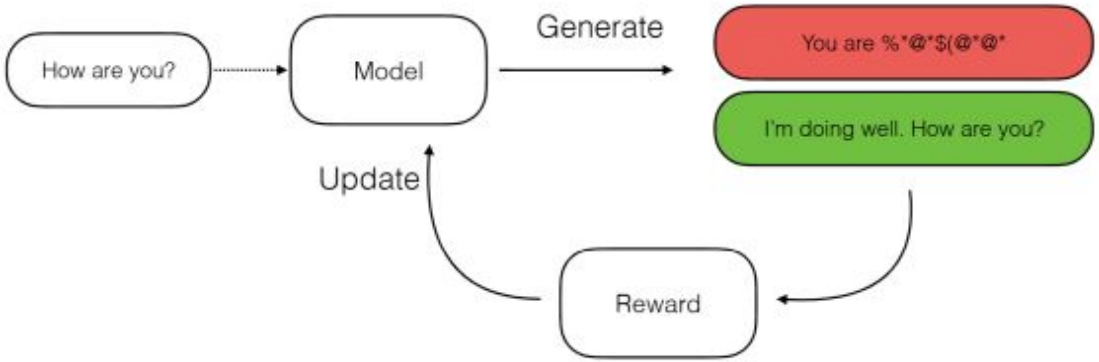
- Key difference 1: The task criteria is now directly optimized via the reward

Reinforcement Learning



- Key difference 2: Data is generated by the model, and a reward tells us howto use the data for training

Reinforcement Learning

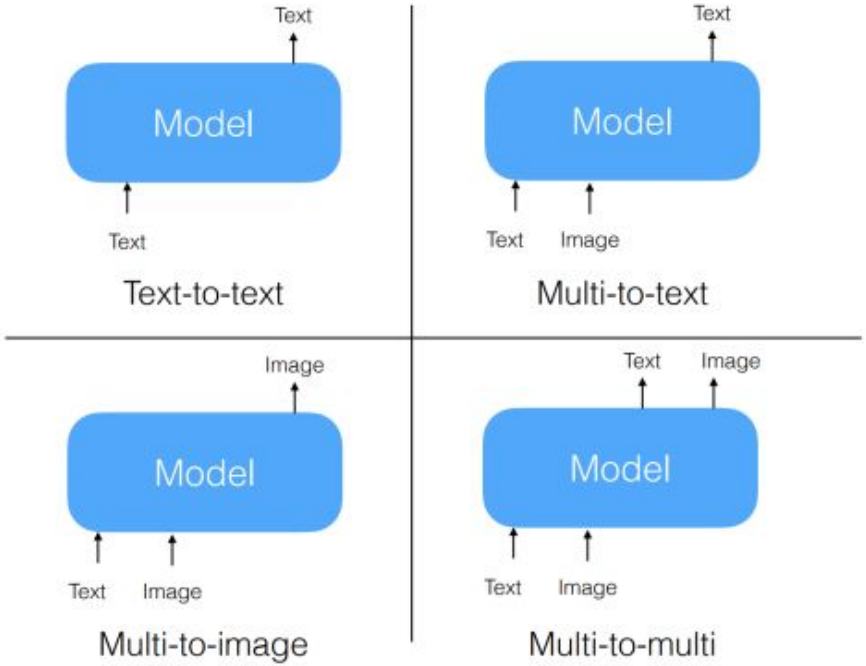


- Key difference 3: Model generations are now in the learning loop, so test-time better resembles training time to use the data for training

Reinforcement Learning

- Reward functions
- Optimizing reward functions
- Reinforcement learning with Human Feedback

Multimodal Modeling



Example: Multi-to-Image

Generated by ChatGPT using Ghibli style.

Project discussion

- How to improve your project with a LLM？
  - e.g., How to solve your problems by using LLMs.